

Joshua Yuen

Electrical Engineering and Physics Student | Photonic Systems &
Optical Computing Researcher | Interdisciplinary Systems Advocate

Boston, MA
(302) 688 9416
yuen.jo@northeastern.edu

PROFESSIONAL EXPERIENCE

Microscale Acoustic and Photonic Systems (MAPS) Lab, Boston, MA— Undergrad Researcher | Nov 2025 – Present

- Developing shot-noise limited Michelson Interferometer to allow for better characterization of microelectromechanical systems (MEMS).
- Designed & assembled high-magnification optical system to assist in the characterization of MEMS devices in a cryogenic vacuum probe station
- Updated and redesigned MATLAB program which calculates displacement of MEMS devices in a Michelson Interferometer under the quadrature condition using the signal-to-noise ratio calculated using experimental parameters from a spectrum analyzer, photodetector, and an oscilloscope.

EECE 2160: Embedded Design Course Assistant, Boston, MA— Hourly Course Assistant (HCA) | Sep 2026 – Present

- Serving as an HCA for two sections of the EECE Fundamental *Embedded Design: Enabling Robotics*
- Specific deliverables include 10 hrs/wk of grading coursework, assisting students with labs, and hosting office hours to facilitate learning in a more personal and interactive environment.

NEU Satellite Labs (NEUSL), Boston, MA— Electrical Team Deputy | Dec 2025 – May 2026; Solar Array Electrical Lead | Sep 2025 – May 2026

- Worked as second-in-command under NEU Satellite Labs Electrical Team Lead to: coordinate and assign electrical projects, serve as a resource for club members seeking assistance with said projects, and instruct new members through the creation of a set onboarding curriculum.

Simply Code (Nonprofit), Philadelphia, PA— Chief Technology Officer | Aug 2024 – Feb 2025

- Developed infrastructure to support mission and operations, including building and launching the organization's first website using WordPress.
- Coordinated volunteers and led the Technology Department, overseeing the planning, execution, and delivery of key technology projects, including partnerships with other organizations in the region.

Drexel University Materials Science Summer Institute, Philadelphia, PA—Lab Student | Aug 2024 – Feb 2025

- Electrospun carbon nanofibers to study how varying electric field strength during the creation process affected fiber diameter & analyzed fibers using scanning electron microscopy to determine the relationship between variables.

PROJECTS & PUBLICATIONS

Project: IMU-Based Hand Gesture Interface for 3D Design and Modeling | Aug 2025 – Present

- Developing a wearable glove-based input system using 9-axis IMU sensors to enable real-time hand gesture control and object manipulation in Blender 3D creation software
- Designing the software-hardware pipeline for motion tracking, gesture interpretation, and CAD integration.
- Exploring shape memory alloy-based haptic feedback to provide tactile responses to virtual object interactions.

Project: Deployable Solar Array for Small Satellites | Sep 2025 – May 2026

- Researching and developing open-source deployable solar arrays for CubeSat power generation.
- Investigated power budgets and researched possible circuit schematics for integrating sensing arrays.
- Awarded PEAK Base Camp Grant from Northeastern University to further project development

Publication: Hide and Seek: A Multimodal Approach to Human Detection | Jul 2024

- Co-authored research paper, "Hide and Seek: A Multimodal Approach to Human Detection" under faculty mentorship which investigated the results of integrating IR sensing, motion profiling, and mmWave sensing into an open source facial recognition software with the goal of increasing accuracy in detecting occluded faces and darker skin tones. Implemented a machine learning model to interpret results.
- Published in the 39th Volume of the Journal of the Pennsylvania Governor's School for the Sciences.

EDUCATION

Northeastern University, Boston, MA— BS in Electrical Engineering and Physics | Aug 2025 – Apr 2029 (Expected)

- National Merit Scholar (Sponsored by Northeastern University).
- GPA: 3.968/4.00 | Dean's List (All Semesters)
- Awarded Gerald F. Tonks Endowed Scholarship (2025-26)
- John Martinson Honors Program: Honors Self-Directed Learner Badge, Honors Impact Badge, and Honors Community Engagement Badge.
- Member of Engineering Honors Society Tau Beta Pi & Electrical & Computer Engineering Honors Society IEEE-Eta Kappa Nu.

Delaware County Christian School (DC), Newtown Square, PA— High School Diploma | Sep 2021 – May 2025

- Graduated Highest Honor Roll. GPA: 5.26 / 5.0 scale. Co-Salutatorian.
- Qualified for the VEX Robotics World Championship (2025); competed as one of a few top-tier teams representing the state of Pennsylvania at the international level in Dallas, TX.
- Co-created and taught DC's 1st accredited engineering class; automated a water dechlorination tool for the water utility company, Aqua.

Pennsylvania Governor's School for the Sciences, Carnegie Mellon University, PA | Summer 2024

- Selected as one of 72 students statewide for an intensive 5-week STEM program that studied topics including Discrete Mathematics, Special Relativity and Genome Editing.

Skills

- **Circuits:** KiCad, Soldering & Circuit Assembly, Microcontrollers, Embedded Design, Raspberry Pi. **Programming:** Python, C++, MATLAB, LaTeX.